Virtual-gate matrix — Test split
4-5-6-7-8_rays_40-50-60_points_500_samples

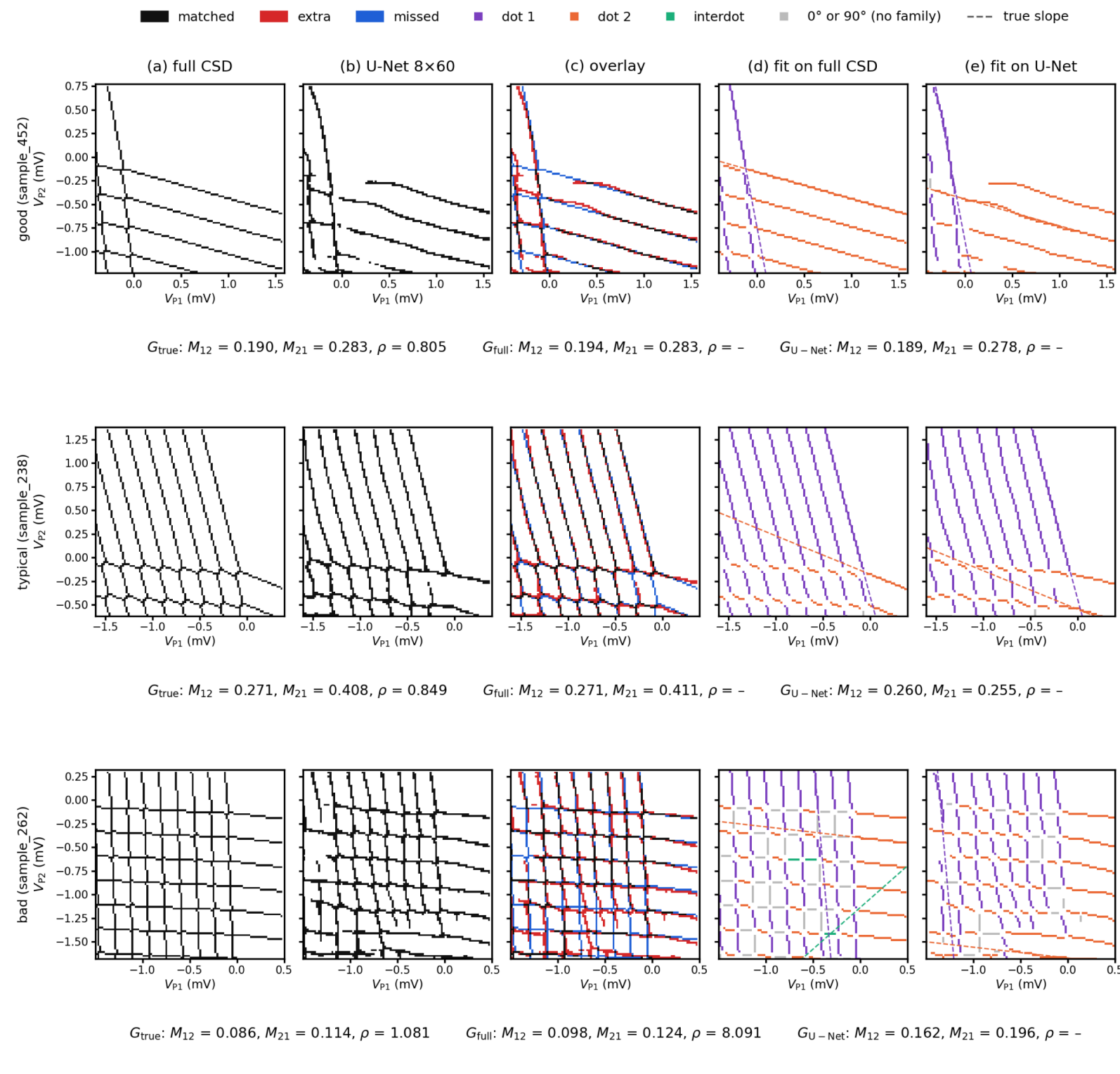
method	rays × points	coverage %	dθ1 (°)	dθ2 (°)	dθ_int (°)	M12 rel. err. (%)	M21 rel. err. (%)	rho rel. err. (%)	interdot found	failed
full CSD	—	100.00	0.09 ± 0.26 [0.06]	-0.02 ± 0.86 [0.04]	-10.86 ± 31.16 [28.27]	1.15 ± 3.40 [0.53]	1.21 ± 8.72 [0.36]	270.56 ± 369.65 [39.58]	8/50	0/50
U-Net	4 × 40	1.57	-3.36 ± 9.03 [3.26]	1.79 ± 8.09 [2.07]	-6.83 ± 33.49 [32.36]	-3.11 ± 54.82 [29.68]	4.93 ± 58.21 [19.55]	90.90 ± 189.19 [65.61]	35/50	0/50
U-Net	4 × 50	1.95	-2.31 ± 6.66 [3.39]	0.87 ± 8.74 [2.66]	-7.28 ± 34.61 [33.00]	3.06 ± 54.44 [31.39]	12.45 ± 59.20 [22.69]	97.03 ± 169.03 [71.77]	34/50	0/50
U-Net	5 × 40	1.94	-5.53 ± 8.81 [3.58]	3.04 ± 8.55 [3.09]	0.12 ± 32.21 [30.26]	-17.95 ± 49.51 [26.04]	-0.15 ± 54.81 [23.84]	50.66 ± 136.92 [72.05]	36/50	0/50
U-Net	4 × 60	2.35	-3.05 ± 8.45 [3.46]	2.33 ± 7.24 [2.83]	-7.14 ± 32.86 [34.88]	1.05 ± 51.68 [31.10]	2.32 ± 49.37 [27.71]	114.57 ± 192.65 [80.03]	35/50	0/50
U-Net	6 × 40	2.33	-1.34 ± 9.09 [2.92]	1.27 ± 6.43 [3.19]	-10.17 ± 34.95 [38.00]	9.33 ± 64.24 [23.21]	6.92 ± 44.89 [24.85]	150.35 ± 208.34 [80.20]	27/50	0/50
U-Net	5 × 50	2.44	-2.01 ± 11.18 [3.10]	3.28 ± 7.97 [2.60]	-8.94 ± 30.45 [27.60]	13.71 ± 117.41 [25.87]	-3.30 ± 45.46 [20.32]	133.15 ± 188.96 [73.25]	30/50	0/50
U-Net	7 × 40	2.72	-1.72 ± 5.33 [2.38]	1.01 ± 6.69 [2.56]	-9.46 ± 28.78 [29.16]	0.44 ± 50.08 [15.60]	10.03 ± 45.09 [22.87]	92.09 ± 170.85 [60.28]	24/50	0/50
U-Net	5 × 60	2.96	-3.09 ± 7.58 [2.23]	2.46 ± 7.69 [2.25]	-21.26 ± 29.53 [38.00]	-3.37 ± 49.98 [18.76]	1.65 ± 43.64 [18.31]	186.92 ± 226.94 [159.03]	34/50	0/50
U-Net	6 × 50	2.94	-4.81 ± 8.93 [2.96]	3.84 ± 8.05 [3.62]	2.72 ± 29.98 [30.93]	-12.36 ± 41.68 [26.45]	-5.05 ± 54.43 [31.87]	47.25 ± 155.36 [59.77]	43/50	0/50
U-Net	8 × 40	3.11	-3.29 ± 8.04 [2.17]	1.63 ± 6.52 [1.80]	-8.55 ± 33.64 [36.37]	-4.53 ± 47.03 [17.75]	1.15 ± 33.81 [17.27]	144.34 ± 225.83 [91.95]	20/50	1/50
U-Net	7 × 50	3.42	-4.28 ± 8.62 [3.05]	0.52 ± 5.87 [1.84]	-2.79 ± 30.95 [29.94]	-9.12 ± 46.74 [25.19]	10.59 ± 43.10 [17.77]	61.24 ± 157.28 [66.17]	36/50	0/50
U-Net	6 × 60	3.52	-3.37 ± 6.64 [2.45]	1.12 ± 5.02 [2.09]	2.00 ± 32.85 [34.56]	-6.53 ± 43.12 [21.83]	3.57 ± 38.51 [15.23]	51.84 ± 151.56 [68.66]	24/50	0/50
U-Net	8 × 50	3.88	-3.26 ± 7.64 [2.07]	2.53 ± 6.73 [1.24]	-4.48 ± 34.38 [33.74]	-6.40 ± 46.36 [16.23]	-2.64 ± 41.62 [12.98]	107.11 ± 199.29 [71.35]	18/50	0/50
U-Net	7 × 60	4.10	-2.32 ± 8.28 [2.28]	1.73 ± 6.67 [1.83]	-6.53 ± 33.79 [34.22]	-1.55 ± 45.95 [17.25]	2.43 ± 38.75 [20.49]	99.81 ± 162.47 [86.62]	27/50	0/50
U-Net	8 × 60	4.66	-1.46 ± 4.44 [1.64]	1.66 ± 6.63 [1.72]	-14.05 ± 30.43 [32.53]	-0.31 ± 31.89 [15.55]	2.06 ± 36.07 [18.56]	160.26 ± 239.09 [73.25]	20/50	0/50

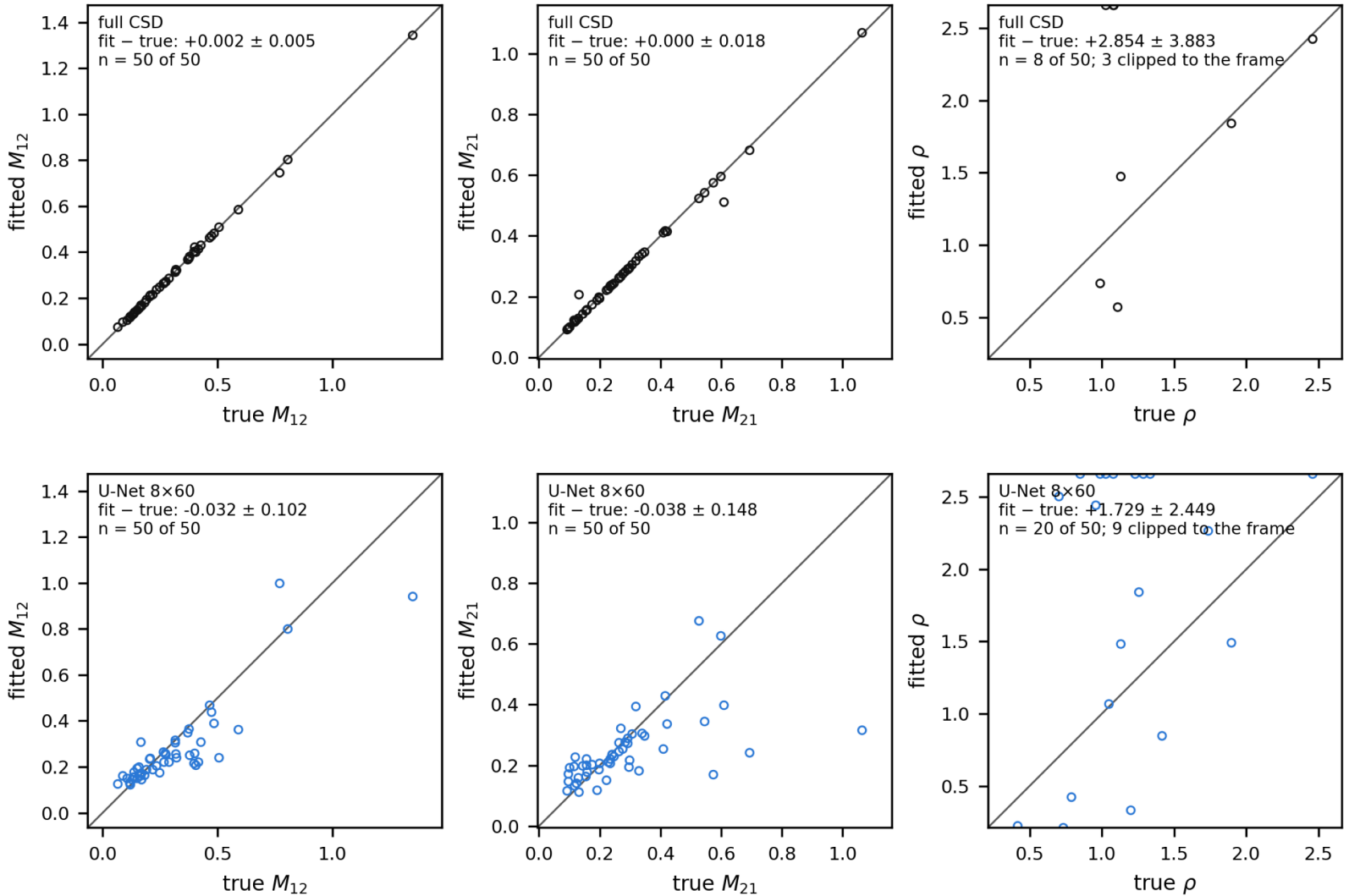
Each error cell: mean ± std over devices where the family was found, median |error| in brackets. 'failed': devices with no dot-1 or no dot-2 piece. rho only where an interdot family was found. The U-Nets were given 1.57–4.66 % of the grid (4 × 40 to 8 × 60 rays × points). On the full CSD the same line fit recovers M12 and M21 with a median relative error of 0.53 % and 0.36 % (41/50 and 44/50 devices within 0.005 of the true value): the floor of the fit itself on 100 × 100 px maps. From the U-Net maps the median relative error of M12 falls from 29.7 % at 1.57 % coverage to 15.6 % at 4.66 %, and that of M21 from 19.6 % to 18.6 %; at 8 × 60 (4.66 %) it is 15.6 % and 18.6 %. The U-Net error is not uniform: at 8 × 60 the devices with a true coupling above 0.4 are underestimated (median -27 % for M12, n = 11; -36 % for M21, n = 10), while those below 0.4 are nearly unbiased (+0.7 % and -0.4 %). The interdot slope, and so rho, is not recovered: interdot segments are a few pixels long, the full CSD yields an interdot family on only 8/50 devices (median |dθ_int| 28.3°, median |rho error| 40 %), and the 18–43 devices per U-Net on which an interdot family is found have median |dθ_int| of 28–38°, i.e. spurious pieces. rho is therefore reported but not claimed. A dot family was missing on 0/50 devices for the full CSD and on at most 1/50 for any U-Net.

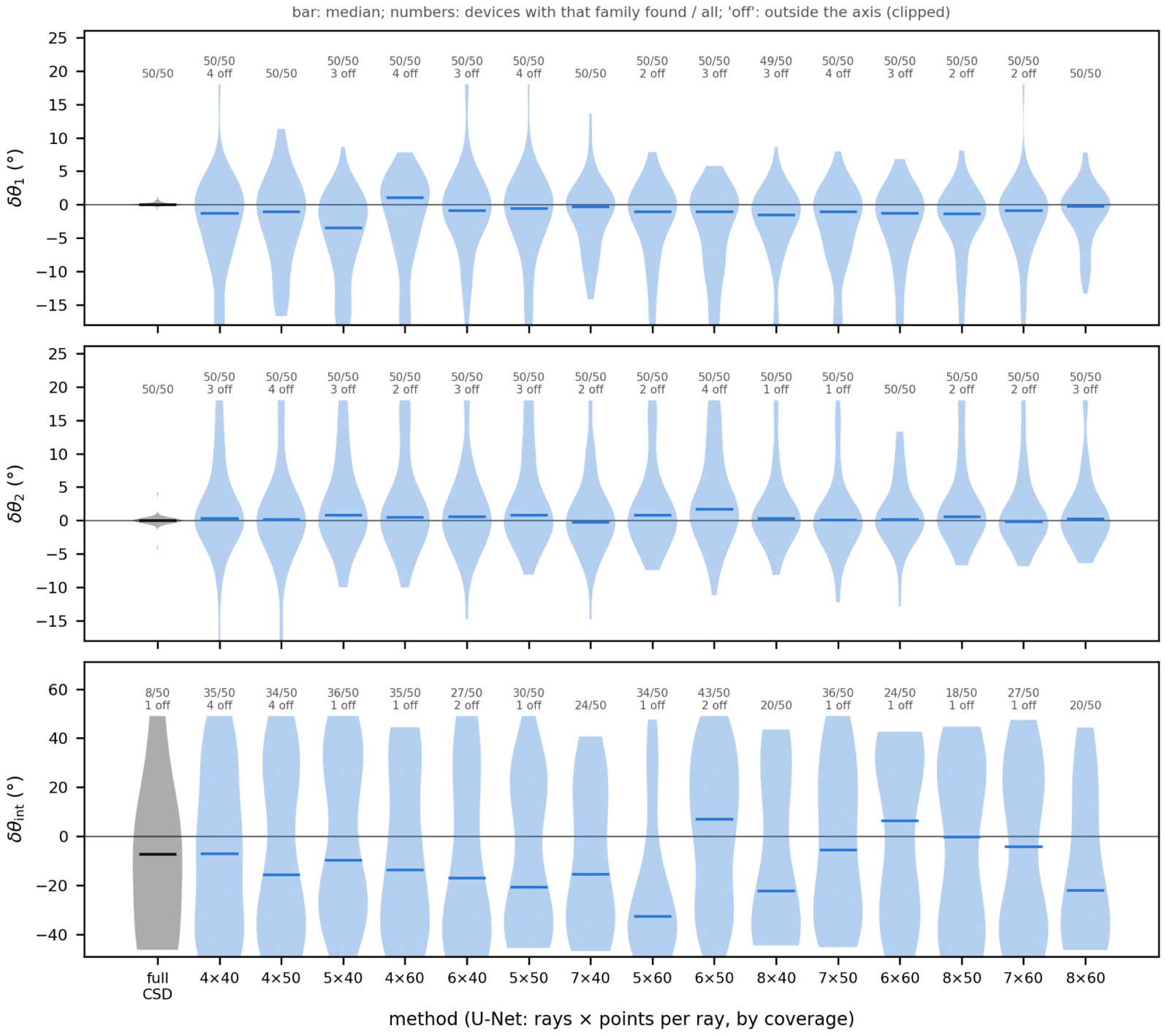
Virtual-gate matrix — Validation split (chose the thresholds; not held out)
4-5-6-7-8_rays_40-50-60_points_500_samples

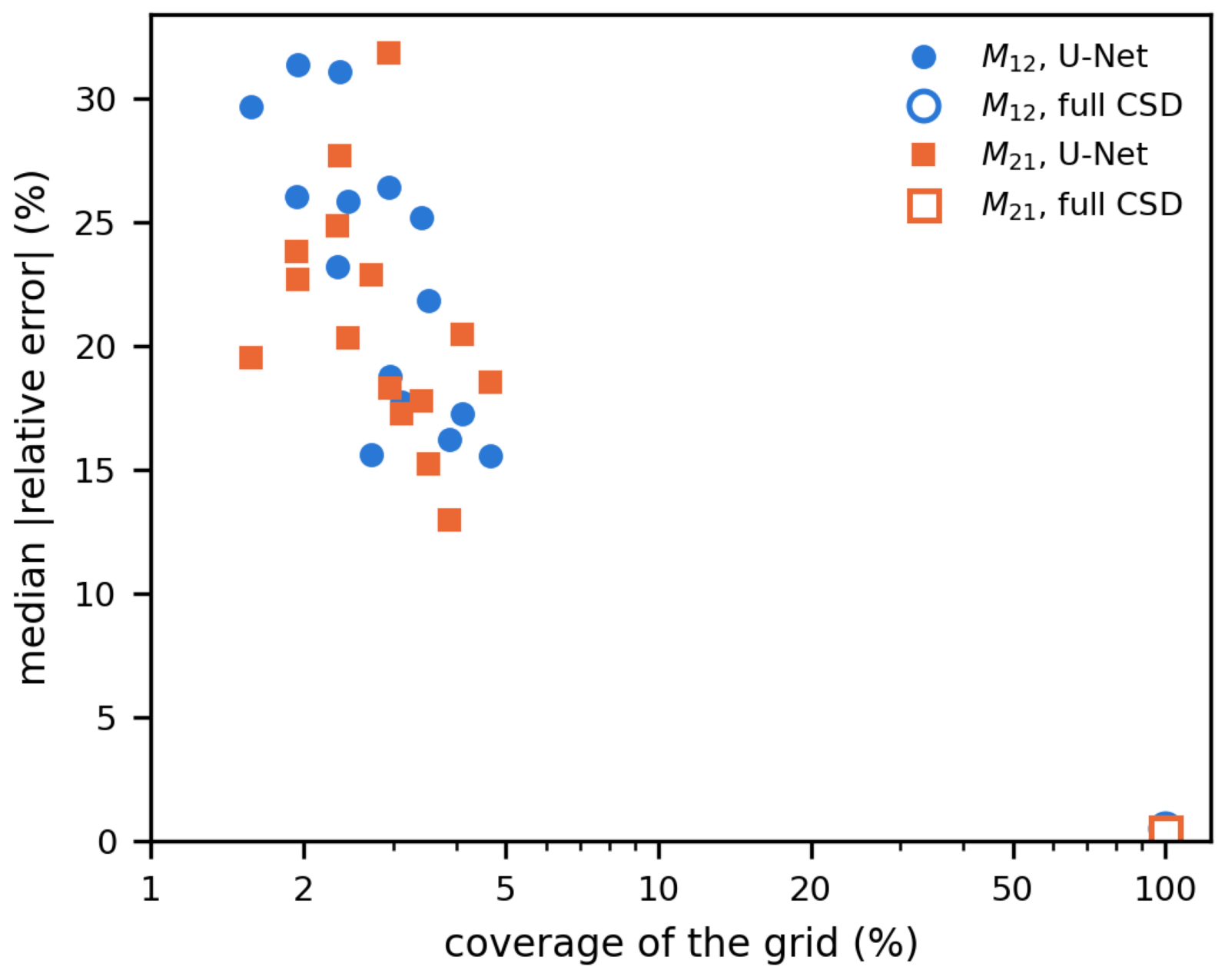
method	rays × points	coverage %	dθ1 (°)	dθ2 (°)	dθ_int (°)	M12 rel. err. (%)	M21 rel. err. (%)	rho rel. err. (%)	interdot found	failed
full CSD	—	100.00	0.10 ± 0.76 [0.05]	-0.10 ± 0.68 [0.04]	-5.56 ± 16.08 [4.07]	1.11 ± 7.03 [0.30]	1.11 ± 7.26 [0.31]	80.21 ± 192.48 [9.52]	16/75	0/75
U-Net	4 × 40	1.57	-9.09 ± 12.92 [4.05]	5.51 ± 11.55 [2.99]	-4.62 ± 32.55 [32.15]	-26.31 ± 39.63 [31.13]	-9.31 ± 46.48 [26.28]	91.55 ± 180.93 [73.31]	44/75	0/75
U-Net	4 × 50	1.95	-6.68 ± 12.27 [4.39]	3.38 ± 10.87 [2.75]	-11.47 ± 33.45 [35.97]	-14.57 ± 44.01 [30.59]	5.98 ± 59.30 [27.28]	125.92 ± 186.02 [75.73]	44/75	0/75
U-Net	5 × 40	1.94	-9.25 ± 12.23 [3.75]	4.13 ± 10.31 [2.21]	-6.63 ± 30.55 [30.04]	-31.73 ± 33.01 [30.23]	-3.14 ± 44.84 [19.49]	89.70 ± 184.82 [78.34]	56/75	0/75
U-Net	4 × 60	2.35	-6.57 ± 11.37 [3.97]	4.07 ± 11.04 [2.66]	-7.87 ± 29.64 [26.72]	-14.33 ± 41.21 [25.92]	1.60 ± 57.21 [30.28]	84.65 ± 158.20 [55.97]	53/75	0/75
U-Net	6 × 40	2.33	-4.99 ± 10.40 [2.60]	3.05 ± 10.33 [1.93]	-13.86 ± 31.45 [33.94]	-10.32 ± 34.67 [21.06]	5.37 ± 55.65 [13.81]	113.48 ± 153.74 [89.09]	40/75	0/75
U-Net	5 × 50	2.44	-6.98 ± 11.70 [2.42]	4.65 ± 12.04 [2.55]	-7.16 ± 35.43 [35.71]	-17.30 ± 37.78 [23.10]	-3.19 ± 49.67 [20.89]	117.67 ± 203.22 [80.44]	49/75	0/75
U-Net	7 × 40	2.72	-5.28 ± 9.30 [2.31]	3.34 ± 10.60 [2.06]	-15.14 ± 30.82 [32.22]	-14.71 ± 30.91 [18.78]	3.49 ± 47.83 [17.73]	133.94 ± 176.27 [105.32]	39/75	0/75
U-Net	5 × 60	2.96	-6.99 ± 11.19 [2.82]	4.37 ± 11.46 [3.16]	-9.06 ± 31.47 [32.56]	-18.74 ± 34.40 [21.35]	-1.92 ± 47.95 [22.65]	128.04 ± 250.02 [72.26]	47/75	0/75
U-Net	6 × 50	2.94	-9.04 ± 12.57 [3.81]	6.15 ± 13.12 [2.33]	5.12 ± 29.71 [26.69]	-27.48 ± 39.21 [35.66]	-11.82 ± 45.23 [25.56]	39.72 ± 174.97 [65.31]	68/75	0/75
U-Net	8 × 40	3.11	-5.29 ± 9.15 [2.71]	3.90 ± 10.65 [2.19]	-10.62 ± 35.60 [38.20]	-15.32 ± 30.19 [17.99]	-2.62 ± 45.52 [18.06]	174.64 ± 246.37 [121.69]	31/75	0/75
U-Net	7 × 50	3.42	-7.57 ± 11.48 [3.56]	2.45 ± 10.00 [1.65]	3.25 ± 32.00 [30.09]	-21.46 ± 36.40 [23.17]	4.85 ± 43.71 [14.93]	43.53 ± 150.62 [67.76]	54/75	0/75
U-Net	6 × 60	3.52	-7.05 ± 11.49 [3.11]	4.16 ± 10.63 [1.81]	-11.77 ± 36.78 [39.49]	-19.05 ± 36.35 [25.07]	-4.50 ± 41.33 [16.38]	172.88 ± 277.47 [154.04]	33/75	0/75
U-Net	8 × 50	3.88	-4.07 ± 8.65 [2.38]	2.99 ± 9.84 [1.74]	-6.44 ± 33.24 [32.63]	-11.04 ± 33.31 [18.90]	1.84 ± 63.18 [14.23]	117.57 ± 244.01 [71.55]	32/75	0/75
U-Net	7 × 60	4.10	-5.73 ± 10.57 [1.90]	3.05 ± 8.83 [1.66]	-12.34 ± 34.00 [36.85]	-14.83 ± 31.86 [14.11]	-0.89 ± 40.50 [15.97]	128.15 ± 170.60 [134.12]	35/75	0/75
U-Net	8 × 60	4.66	-4.31 ± 9.43 [1.93]	3.67 ± 9.65 [2.45]	-15.53 ± 29.30 [30.29]	-9.38 ± 31.21 [11.19]	-4.19 ± 41.10 [18.23]	167.14 ± 276.84 [88.78]	25/75	0/75

Each error cell: mean ± std over devices where the family was found, median |error| in brackets. 'failed': devices with no dot-1 or no dot-2 piece. rho only where an interdot family was found.


matched
extra
missed
dot 1
dot 2
interdot
0° or 90° (no family)
true slope
(a) full CSD
(b) U-Net 8×60
(c) overlay
(d) fit on full CSD
(e) fit on U-Net
good (sample_452)
typical (sample_238)
bad (sample_262)
V_{P1} (mV)
V_{P2} (mV)
G_{true}: M_{12} = 0.190, M_{21} = 0.283, ρ = 0.805
G_{full}: M_{12} = 0.194, M_{21} = 0.283, ρ = –
G_{U-Net}: M_{12} = 0.189, M_{21} = 0.278, ρ = –
G_{true}: M_{12} = 0.271, M_{21} = 0.408, ρ = 0.849
G_{full}: M_{12} = 0.271, M_{21} = 0.411, ρ = –
G_{U-Net}: M_{12} = 0.260, M_{21} = 0.255, ρ = –
G_{true}: M_{12} = 0.086, M_{21} = 0.114, ρ = 1.081
G_{full}: M_{12} = 0.098, M_{21} = 0.124, ρ = 8.091
G_{U-Net}: M_{12} = 0.162, M_{21} = 0.196, ρ = –


full CSD
fit − true: +0.002 ± 0.005
n = 50 of 50
fitted M_{12}
true M_{12}
full CSD
fit − true: +0.000 ± 0.018
n = 50 of 50
fitted M_{21}
true M_{21}
full CSD
fit − true: +2.854 ± 3.883
n = 8 of 50; 3 clipped to the frame
fitted ρ
true ρ
U-Net 8×60
fit − true: -0.032 ± 0.102
n = 50 of 50
fitted M_{12}
true M_{12}
U-Net 8×60
fit − true: -0.038 ± 0.148
n = 50 of 50
fitted M_{21}
true M_{21}
U-Net 8×60
fit − true: +1.729 ± 2.449
n = 20 of 50; 9 clipped to the frame
fitted ρ
true ρ


bar: median; numbers: devices with that family found / all; 'off': outside the axis (clipped)
$\delta\theta_1$ (°)
$\delta\theta_2$ (°)
$\delta\theta_{\mathrm{int}}$ (°)
50/50
50/50 4 off
50/50
50/50 3 off
50/50 4 off
50/50 3 off
50/50 4 off
50/50
50/50 2 off
50/50 3 off
49/50 3 off
50/50 4 off
50/50 3 off
50/50 2 off
50/50 2 off
50/50
50/50
50/50 3 off
50/50 4 off
50/50 3 off
50/50 2 off
50/50 3 off
50/50 3 off
50/50 2 off
50/50 2 off
50/50 4 off
50/50 1 off
50/50 1 off
50/50
50/50 2 off
50/50 2 off
50/50 3 off
8/50 1 off
35/50 4 off
34/50 4 off
36/50 1 off
35/50 1 off
27/50 2 off
30/50 1 off
24/50
34/50 1 off
43/50 2 off
20/50
36/50 1 off
24/50 1 off
18/50 1 off
27/50 1 off
20/50
full CSD
4×40
4×50
5×40
4×60
6×40
5×50
7×40
5×60
6×50
8×40
7×50
6×60
8×50
7×60
8×60
method (U-Net: rays × points per ray, by coverage)


M_{12}, U-Net
M_{12}, full CSD
M_{21}, U-Net
M_{21}, full CSD
median |relative error| (%)
0
5
10
15
20
25
30
1
2
5
10
20
50
100
coverage of the grid (%)